

Problem 3 (Bulgarian solitaire), Cell 2: $D_{\mathcal{B}}$ at triangular n

Problem 3 team

Official setting.

A partition of a positive integer n is a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_s)$ of positive integers with $\lambda_1 + \dots + \lambda_s = n$. The shift $\mathcal{B}(\lambda)$ is the partition whose parts are the positive numbers among $\lambda_1 - 1, \dots, \lambda_s - 1$ together with one extra part equal to s . Call λ cyclic if $\mathcal{B}^i(\lambda) = \lambda$ for some $i \geq 1$, and let $d_{\mathcal{B}}(\lambda) = \min\{i \geq 0 : \mathcal{B}^i(\lambda) \text{ is cyclic}\}$, $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \text{ is a partition of } n\}$. Write $T_k = k(k+1)/2$ and $\delta_k = (k, k-1, \dots, 2, 1)$. Every $n \geq 1$ satisfies $T_{k-1} < n \leq T_k$ for exactly one k , which we call the rank of n .

Section 1 restates these definitions in an equivalent form, proves that $d_{\mathcal{B}}$ is well defined and that the rank exists and is unique (Lemma 1.2, where we also write $n = T_{k-1} + r$, $1 \leq r \leq k$), and fixes further notation.

Official statement of the cell.

Determine $D_{\mathcal{B}}(T_k)$ for every k , with proof of both bounds.

Status. Solved (complete proof; no computation is used for the value of $D_{\mathcal{B}}(T_k)$).

Answer and where it is proved.

- $D_{\mathcal{B}}(T_k) = k^2 - k$ for every $k \geq 1$ (Theorem 7.3). Explicitly, for the smallest cases: $D_{\mathcal{B}}(T_1) = D_{\mathcal{B}}(1) = 0$, $D_{\mathcal{B}}(T_2) = D_{\mathcal{B}}(3) = 2$ and $D_{\mathcal{B}}(T_3) = D_{\mathcal{B}}(6) = 6$. These are attained by (1) , $(1, 1, 1)$ and $(2, 2, 1, 1)$, and all three are covered by the proofs below. $k = 1, 2$ are done by hand in Theorem 7.2; $k = 3$ is inside the general argument ($k \geq 3$) and the witness family ($k \geq 2$).
- Lower bound: for $k \geq 2$ the partition $\lambda^{[k]} = (k-1, k-1, k-2, k-3, \dots, 2, 1, 1)$ ($\lambda_1 = k-1$, $\lambda_i = k+1-i$ for $2 \leq i \leq k$, $\lambda_{k+1} = 1$) has $d_{\mathcal{B}}(\lambda^{[k]}) = k^2 - k$ (Theorem 6.2, [4, Thm. 3.1]).
- Upper bound: $d_{\mathcal{B}}(\lambda) \leq k^2 - k$ for all $\lambda \vdash T_k$ (Theorem 7.2, the Igusa–Etienne theorem [5, 3]). We reproduce, completed in detail, the proof of Griggs–Ho [4, Lemmas 3.3–3.6, Thm. 3.7].
- In addition (not asked): for $k \geq 4$, $\lambda \vdash T_k$ is extremal iff $\mathcal{B}^{k^2-k-2}(\lambda)$ has at least $k+1$ parts (Theorem 8.1).

Sections 1–5 are shared with Cell 3 (and, up to Section 4, with Cells 4 and 5), and are repeated so that this document is self-contained.

1 Setting and notation

Throughout, $T_m = m(m+1)/2$ for integers $m \geq 0$, and $[P] \in \{0, 1\}$ denotes the truth value of a statement P . A *partition* λ of $n \geq 1$ (written $\lambda \vdash n$) is a finite non-increasing sequence of positive integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_\ell$ with sum n ; $\ell = \ell(\lambda)$ is its number of parts, and we put $\lambda_i = 0$ for $i > \ell(\lambda)$. Partitions are identified with multisets of positive integers.

Definition 1.1 (Bulgarian solitaire). For $\lambda \vdash n$ let $\mathcal{B}(\lambda)$ be the partition whose parts are $\ell(\lambda)$ together with the numbers $\lambda_i - 1$ for those $i \leq \ell(\lambda)$ with $\lambda_i \geq 2$. Since $\ell(\lambda) + \sum_i (\lambda_i - 1) = n$, we have $\mathcal{B}(\lambda) \vdash n$. Put $\mathcal{B}^0 = \text{id}$ and $\mathcal{B}^{i+1} = \mathcal{B} \circ \mathcal{B}^i$. A partition μ is *cyclic* if $\mathcal{B}^P(\mu) = \mu$ for some $P \geq 1$. For $\lambda \vdash n$, $d_{\mathcal{B}}(\lambda)$ is the least $i \geq 0$ such that $\mathcal{B}^i(\lambda)$ is cyclic, and $D_{\mathcal{B}}(n) = \max\{d_{\mathcal{B}}(\lambda) : \lambda \vdash n\}$. A partition $\lambda \vdash n$ with $d_{\mathcal{B}}(\lambda) = D_{\mathcal{B}}(n)$ is called *extremal*.

$d_{\mathcal{B}}(\lambda)$ is well defined: n has finitely many partitions, so $\mathcal{B}^a(\lambda) = \mathcal{B}^{a+P}(\lambda)$ for some $a \geq 0$, $P \geq 1$, and then $\mathcal{B}^a(\lambda)$ is cyclic. If μ is cyclic with $\mathcal{B}^P(\mu) = \mu$, then $\mathcal{B}^P(\mathcal{B}(\mu)) = \mathcal{B}(\mu)$, so $\mathcal{B}(\mu)$ is cyclic. Hence $\mathcal{B}^i(\lambda)$ is cyclic for every $i \geq d_{\mathcal{B}}(\lambda)$ and for no $i < d_{\mathcal{B}}(\lambda)$.

Lemma 1.2 (rank). *Every $n \geq 1$ can be written in exactly one way as $n = T_{k-1} + r$ with integers $k \geq 1$ and $1 \leq r \leq k$. We call k the rank of n ; n is triangular iff $r = k$ (then $n = T_k$).*

Proof. $T_0 = 0 < T_1 < T_2 < \dots$ is strictly increasing and unbounded, so there is exactly one $k \geq 1$ with $T_{k-1} < n \leq T_k$; then $r = n - T_{k-1}$ satisfies $1 \leq r \leq T_k - T_{k-1} = k$. Conversely $n = T_{k-1} + r$ with $1 \leq r \leq k$ forces $T_{k-1} < n \leq T_k$, i.e. this k . If $n = T_m$ ($m \geq 1$), then $T_{m-1} < n \leq T_m$, so $k = m$ and $r = T_m - T_{m-1} = m = k$; conversely $r = k$ gives $n = T_k$. $\square$

2 Diagrams, diagonals, rotation, and Brandt's theorem

This section follows Griggs–Ho [4, §2] (the idea goes back to Etienne [3]); we make their “left shifting decreases the diagonal index” precise with an explicit potential Φ .

Let $\mathbb{P} = \mathbb{Z}_{>0} \times \mathbb{Z}_{>0}$; the *cell* $(i, j) \in \mathbb{P}$ has *row* i and *column* j . The *diagram* of λ is

$$Y(\lambda) = \{(i, j) \in \mathbb{P} : 1 \leq j \leq \ell(\lambda), 1 \leq i \leq \lambda_j\},$$

so column j has λ_j cells (columns are parts, as in [4]). $Y(\lambda)$ determines λ , since $\lambda_j = \#\{i : (i, j) \in Y(\lambda)\}$. Row 1 of $Y(\lambda)$ is $\{(1, j) : j \leq \ell(\lambda)\}$ and column 1 has λ_1 cells; in particular

$$(1, w+1) \in Y(\lambda) \implies (1, w) \in Y(\lambda) \quad (w \geq 1). \quad (1)$$

For a finite sequence $a = (a_1, \dots, a_m)$ of non-negative integers let $X_a = \{(i, j) : 1 \leq j \leq m, 1 \leq i \leq a_j\}$. If a is non-increasing, then $X_a = Y(\nu)$ where ν is the partition formed by the non-zero entries of a .

For $w \geq 1$ the w -th *diagonal* is $D_w = \{(i, j) \in \mathbb{P} : i + j = w + 1\}$; it has the w cells $d_w(j) = (w + 1 - j, j)$, $1 \leq j \leq w$ ($d_w(j)$ is the cell of D_w in column j). $\mathbb{P}$ is the disjoint union of the D_w . Put $\Delta_0 = \emptyset$ and $\Delta_m = D_1 \cup \dots \cup D_m$ for $m \geq 1$. Column j of Δ_m consists of the cells (i, j) with $i \leq m + 1 - j$, so $\Delta_m = Y(\delta_m)$ with $\delta_m = (m, m - 1, \dots, 1)$, and $|\Delta_m| = T_m$.

Define $\rho : \mathbb{P} \rightarrow \mathbb{P}$ by $\rho(i, j) = (i - 1, j + 1)$ if $i \geq 2$ and $\rho(1, j) = (j, 1)$.

Lemma 2.1. (a) $\rho(d_w(j)) = d_w(j + 1)$ for $1 \leq j < w$ and $\rho(d_w(w)) = d_w(1)$. Hence ρ maps each D_w bijectively onto itself, ρ is a bijection of $\mathbb{P}$, and for $m \geq 0$, $\rho^m(d_w(j)) = d_w(j')$ where $1 \leq j' \leq w$ and $j' \equiv j + m \pmod{w}$. In particular ρ^w is the identity on D_w . (b) ρ preserves the quantity $i + j$ of a cell.

Proof. (a) For $j < w$ the cell $d_w(j) = (w+1-j, j)$ has row $w+1-j \geq 2$, so $\rho(d_w(j)) = (w-j, j+1) = d_w(j+1)$; and $\rho(d_w(w)) = \rho(1, w) = (w, 1) = d_w(1)$. So ρ acts on D_w as the cyclic shift $j \mapsto j+1$ of column indices modulo w ; the rest follows. (b) $(i-1) + (j+1) = i+j$ and $j+1 = 1+j$. $\square$

Lemma 2.2 (one move). *Let $\lambda \vdash n$, $s = \ell(\lambda)$ and $a = (s, \lambda_1 - 1, \lambda_2 - 1, \dots, \lambda_s - 1)$. Then $\rho(Y(\lambda)) = X_a$, and $Y(\mathcal{B}(\lambda)) = X_{a^*}$, where a^* is the non-increasing rearrangement of a . If $s \geq \lambda_1 - 1$, then a is non-increasing and $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.*

Proof. The cells of row 1 of $Y(\lambda)$ are $(1, j)$, $1 \leq j \leq s$; ρ sends them to $(j, 1)$, i.e. exactly onto the cells $(1, 1), \dots, (s, 1)$ of column 1. A cell $(i, j) \in Y(\lambda)$ with $i \geq 2$ satisfies $2 \leq i \leq \lambda_j$ and is sent to $(i-1, j+1)$; for fixed j these images are exactly the cells $(1, j+1), \dots, (\lambda_j - 1, j+1)$. Hence $\rho(Y(\lambda)) = X_a$. The parts of $\mathcal{B}(\lambda)$ are the non-zero entries of a , so $Y(\mathcal{B}(\lambda)) = X_{a^*}$. The entries of a after the first are non-increasing because λ is; so a is non-increasing iff $s \geq \lambda_1 - 1$, and then $a^* = a$. $\square$

For a finite set $X \subseteq \mathbb{P}$ put $\Phi(X) = \sum_{(i,j) \in X} (i+j)$.

Lemma 2.3. *For every finite sequence a of non-negative integers, $\Phi(X_{a^*}) \leq \Phi(X_a)$, with equality iff a is non-increasing.*

Proof. $\Phi(X_a) = \sum_j \sum_{i=1}^{a_j} (i+j) = \sum_j \binom{a_j+1}{2} + F(a)$ with $F(a) = \sum_j j a_j$. The first sum does not change when a is permuted. If $a_j < a_{j+1}$, exchanging these two entries changes F by $j a_{j+1} + (j+1) a_j - j a_j - (j+1) a_{j+1} = a_j - a_{j+1} < 0$. Starting from a , repeatedly exchange some adjacent pair with $a_j < a_{j+1}$. Each exchange lowers the non-negative integer F , so the process stops, and it stops at a sequence with no adjacent ascent, i.e. at the non-increasing rearrangement a^* (which is unique). If a is not non-increasing at least one exchange happens, so $F(a^*) < F(a)$; if a is non-increasing then $a = a^*$. $\square$

Corollary 2.4. *For $\lambda \vdash n$: $\Phi(Y(\mathcal{B}(\lambda))) \leq \Phi(Y(\lambda))$, with equality iff $\ell(\lambda) \geq \lambda_1 - 1$; in the case of equality $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$.*

Proof. By Lemma 2.1, ρ is injective and preserves $i+j$, so $\Phi(\rho(X)) = \Phi(X)$. With a as in Lemma 2.2: $\Phi(Y(\lambda)) = \Phi(X_a) \geq \Phi(X_{a^*}) = \Phi(Y(\mathcal{B}(\lambda)))$, with equality iff a is non-increasing (Lemma 2.3), i.e. iff $\ell(\lambda) \geq \lambda_1 - 1$, and then Lemma 2.2 gives $Y(\mathcal{B}(\lambda)) = \rho(Y(\lambda))$. $\square$

For $k \geq 1$ and $E \subseteq D_k$ let λ_E be the partition whose j -th part, $1 \leq j \leq k$, is $k-j + [d_k(j) \in E]$ (zero parts omitted). The sequence $(k-j + \varepsilon_j)_{j=1}^k$ with $\varepsilon_j \in \{0, 1\}$ is non-increasing (consecutive differences are $1 + \varepsilon_j - \varepsilon_{j+1} \geq 0$), so λ_E is a partition, $|\lambda_E| = T_{k-1} + |E|$, and $Y(\lambda_E) = \Delta_{k-1} \cup E$ (column j consists of the $k-j$ cells of Δ_{k-1} and, if $d_k(j) \in E$, the cell $d_k(j) = (k+1-j, j)$). Writing $e_{k-j} = [d_k(j) \in E]$ we get $\lambda_E = (k-1 + e_{k-1}, k-2 + e_{k-2}, \dots, 1 + e_1, e_0)$.

Theorem 2.5 (Brandt [2]; see also [1, 3, 4]). *Let $n = T_{k-1} + r$ with $1 \leq r \leq k$. A partition of n is cyclic if and only if it equals λ_E for some $E \subseteq D_k$ with $|E| = r$; equivalently, iff it has the form $(k-1 + e_{k-1}, \dots, 1 + e_1, e_0)$ with $e_i \in \{0, 1\}$ and $\sum_i e_i = r$ (zero parts omitted). Moreover $Y(\mathcal{B}(\lambda_E)) = \rho(Y(\lambda_E))$, i.e. $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and $\mathcal{B}^k(\lambda_E) = \lambda_E$.*

Proof. ($\Leftarrow$ and “moreover”). Let $\mu = \lambda_E$ with $E \subseteq D_k$, $|E| = r$. Row 1 of $Y(\mu) = \Delta_{k-1} \cup E$ consists of the cells $(1, w) = d_w(w)$ with $w \leq k-1$ and possibly $d_k(k)$; so $\ell(\mu) = k-1 + [d_k(k) \in E] \geq k-1$. Column 1 consists of $d_w(1)$ for $w \leq k-1$ and possibly $d_k(1)$; so $\mu_1 = k-1 + [d_k(1) \in E] \leq k$. Thus $\ell(\mu) \geq \mu_1 - 1$ and Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(\Delta_{k-1} \cup \rho(E)) = \Delta_{k-1} \cup \rho(E)$ with $\rho(E) \subseteq D_k$, $|\rho(E)| = r$. Hence $\mathcal{B}(\lambda_E) = \lambda_{\rho(E)}$, and by induction $\mathcal{B}^k(\lambda_E) = \lambda_{\rho^k(E)} = \lambda_E$ because ρ^k is the identity on D_k . So μ is cyclic.

($\Rightarrow$). Let $\lambda \vdash n$ with $\mathcal{B}^P(\lambda) = \lambda$, $P \geq 1$, and write $\lambda^{(m)} = \mathcal{B}^m(\lambda)$. By Corollary 2.4, $\Phi(Y(\lambda^{(0)})) \geq \Phi(Y(\lambda^{(1)})) \geq \dots \geq \Phi(Y(\lambda^{(P)})) = \Phi(Y(\lambda^{(0)}))$, so all these are equalities and $Y(\lambda^{(m+1)}) = \rho(Y(\lambda^{(m)}))$ for $0 \leq m < P$. As $\lambda^{(m+P)} = \lambda^{(m)}$, this holds for all $m \geq 0$, and therefore $Y(\lambda^{(m)}) = \rho^m(Y(\lambda))$ for all $m \geq 0$.

Claim: there is no $w \geq 1$ such that $D_w \not\subseteq Y(\lambda)$ and $D_{w+1} \cap Y(\lambda) \neq \emptyset$. Suppose $d_w(\alpha) \notin Y(\lambda)$ and $d_{w+1}(\beta) \in Y(\lambda)$. Since $\gcd(w, w+1) = 1$, the Chinese remainder theorem gives $m \geq 0$ with $m \equiv w - \alpha \pmod{w}$ and $m \equiv w + 1 - \beta \pmod{w+1}$. By Lemma 2.1, $\rho^m(d_w(\alpha)) = d_w(w) = (1, w)$ and $\rho^m(d_{w+1}(\beta)) = d_{w+1}(w+1) = (1, w+1)$. As ρ^m is injective, $(1, w) \notin \rho^m(Y(\lambda)) = Y(\lambda^{(m)})$ while $(1, w+1) \in Y(\lambda^{(m)})$, contradicting (1).

$Y(\lambda)$ is finite, so some D_w is not contained in it; let $K \geq 1$ be the least such w . Then $\Delta_{K-1} \subseteq Y(\lambda)$. By the claim, $D_{K+1} \cap Y(\lambda) = \emptyset$; then $D_{K+1} \not\subseteq Y(\lambda)$, so again by the claim $D_{K+2} \cap Y(\lambda) = \emptyset$, and inductively $D_w \cap Y(\lambda) = \emptyset$ for all $w > K$. Thus $Y(\lambda) = \Delta_{K-1} \cup E'$ with $E' = Y(\lambda) \cap D_K \neq D_K$; let $r' = |E'| \in \{0, \dots, K-1\}$, so $n = T_{K-1} + r'$ and $Y(\lambda) = Y(\lambda_{E'})$ (with k replaced by K in the definition of $\lambda_{E'}$). If $r' \geq 1$, then $T_{K-1} < n < T_K$, so by Lemma 1.2 $K = k$, $r' = r$, and $\lambda = \lambda_{E'}$. If $r' = 0$, then $K \geq 2$ (as $n \geq 1$) and $n = T_{K-1} = T_{K-2} + (K-1)$, so $k = K-1$, $r = k$, and $Y(\lambda) = \Delta_k = \Delta_{k-1} \cup D_k$, i.e. $\lambda = \lambda_{D_k}$ with $|D_k| = k = r$. $\square$

Corollary 2.6. *Let $n = T_{k-1} + r$, $1 \leq r \leq k$.*

- (a) *If $r = k$ (i.e. $n = T_k$), the only cyclic partition of n is $\delta_k = (k, k-1, \dots, 1)$, and $\mathcal{B}(\delta_k) = \delta_k$.*
- (b) *Every cyclic $\mu \vdash n$ satisfies $Y(\mu) \subseteq \Delta_k$, $\Delta_{k-1} \subseteq Y(\mu)$, $\ell(\mu) \in \{k-1, k\}$ and $\mathcal{B}^k(\mu) = \mu$.*
- (c) *If $\mu \vdash n$ is cyclic, then for every $a \geq 0$: $\sum_{i=a}^{a+k-1} \ell(\mathcal{B}^i(\mu)) = k(k-1) + r$. If moreover $r < k$, both values $k-1$ and k occur among $\ell(\mathcal{B}^a(\mu)), \dots, \ell(\mathcal{B}^{a+k-1}(\mu))$.*

Proof. (a) and (b) are immediate from Theorem 2.5 ($E = D_k$ when $r = k$; $\ell(\lambda_E) = k-1 + [d_k(k) \in E]$ was computed in its proof). (c) Write $\mu = \lambda_E$. Then $\mathcal{B}^i(\mu) = \lambda_{\rho^i(E)}$, so $\ell(\mathcal{B}^i(\mu)) = k-1 + [d_k(k) \in \rho^i(E)] = k-1 + [d_k(j_i) \in E]$, where $d_k(j_i) = \rho^{-i}(d_k(k))$, i.e. $j_i \equiv k-i \pmod{k}$ (Lemma 2.1). As i runs through $a, \dots, a+k-1$, j_i runs through all of $1, \dots, k$ once, so the sum is $k(k-1) + |E| = k(k-1) + r$. If $0 < r < k$, some j_i lies in E and some does not. $\square$

3 Piles and the part-count sequence

This is the “ $\text{diagram}_B(\lambda)$ ” bookkeeping of Griggs–Ho [4, §3], written with explicitly labelled piles.

Fix $\lambda \vdash n$, write $\lambda^{(i)} = \mathcal{B}^i(\lambda)$ and

$$c_i = \ell(\lambda^{(i-1)}) \quad (i \geq 1).$$

We call $c(\lambda) = (c_1, c_2, \dots)$ the *part-count sequence*; clearly $c(\lambda^{(a)}) = (c_{a+1}, c_{a+2}, \dots)$. Every $c_i \geq 1$.

Lemma 3.1 (piles). *For every $i \geq 0$ the multiset of parts of $\lambda^{(i)}$ is*

$$\{\lambda_m - i : 1 \leq m \leq \ell(\lambda), \lambda_m > i\} \uplus \{c_j - (i-j) : 1 \leq j \leq i, c_j > i-j\}.$$

Proof. Induction on i ; $i = 0$ is clear. $\lambda^{(i+1)} = \mathcal{B}(\lambda^{(i)})$ has the part $\ell(\lambda^{(i)}) = c_{i+1} = c_{i+1} - 0$ and the parts of $\lambda^{(i)}$ decreased by one, the zeros being discarded: this is the stated multiset for $i+1$ (the term $j = i+1$ is c_{i+1}). $\square$

We use *labels*: $O_1, \dots, O_{\ell(\lambda)}$ (the original piles) and $N_1, N_2, \dots$ (N_j is the pile created by the j -th move, of size c_j when created). By Lemma 3.1, at time i (i.e. in $\lambda^{(i)}$) the pile O_m has size $\lambda_m - i$ and the pile N_j ($j \leq i$) has size $c_j - (i - j)$, a pile being present iff its size is positive. For $i \geq 1$ let R_i be the set of labels of the piles present at time $i - 1$:

$$O_m \in R_i \iff \lambda_m \geq i, \quad N_j \in R_i \iff j \leq i - 1 \text{ and } c_j \geq i - j. \quad (2)$$

By Lemma 3.1 (with $i - 1$), $|R_i| = \ell(\lambda^{(i-1)}) = c_i$. Let $Z_i = R_i \setminus R_{i+1}$ (the piles of size 1 at time $i - 1$, which disappear in move i).

Lemma 3.2 (basic facts). *For all $i \geq 1$:*

- (a) $N_i \in R_{i+1}$ and $N_i \notin R_i$.
- (b) $R_{i+1} = (R_i \setminus Z_i) \cup \{N_i\}$, hence $c_{i+1} = c_i + 1 - |Z_i|$ and $c_{i+1} \leq c_i + 1$.
- (c) If $c_{i+1} = c_i + 1$ then $Z_i = \emptyset$, i.e. $R_i \subseteq R_{i+1}$. If $c_{i+1} = c_i$ then $|Z_i| = 1$.
- (d) For $j < i$: $N_j \in R_i$ implies $N_j \in R_{i'}$ for $j < i' \leq i$, and $N_j \notin R_i$ implies $N_j \notin R_{i'}$ for all $i' \geq i$. Likewise for the labels O_m .

Proof. (a) $c_i \geq 1 = (i + 1) - i$; and labels N_j in R_i have $j \leq i - 1$. (d) For $i' > j$ the clause $j \leq i' - 1$ in (2) holds automatically, so membership of N_j in $R_{i'}$ is the condition $c_j \geq i' - j$, which is monotone in i' ; for O_m the condition $\lambda_m \geq i'$ is monotone as well. (b) By (d), $R_{i+1} \setminus \{N_i\} \subseteq R_i$, so $R_{i+1} = (R_i \cap R_{i+1}) \cup \{N_i\} = (R_i \setminus Z_i) \cup \{N_i\}$, a disjoint union by (a); take cardinalities. (c) follows from $c_{i+1} = c_i + 1 - |Z_i|$. $\square$

Lemma 3.3 (gap lemma). *Let $1 \leq j < i$. If $N_j \notin R_i$ and $N_{j+1} \in R_{i+1}$, then $c_j = i - j - 1$, $c_{j+1} = i - j$, and $N_{j+1} \notin R_{i+2}$.*

Proof. By (2), $c_j \leq i - j - 1$ and $c_{j+1} \geq (i + 1) - (j + 1) = i - j$. By Lemma 3.2(b), $c_{j+1} \leq c_j + 1 \leq i - j$. Hence both equalities hold, and $c_{j+1} = i - j < (i + 2) - (j + 1)$ gives $N_{j+1} \notin R_{i+2}$. $\square$

Lemma 3.4 (the sequence determines the partition). *For $i \geq 1$, $\#\{m : \lambda_m \geq i\} = c_i - \#\{j : 1 \leq j \leq i - 1, c_j \geq i - j\}$. Consequently, if two partitions have the same part-count sequence, they are equal.*

Proof. The formula is $|R_i| = c_i$ together with (2). The numbers $\#\{m : \lambda_m \geq i\}$, $i \geq 1$, determine λ , since $\lambda_m = \#\{i \geq 1 : \#\{m' : \lambda_{m'} \geq i\} \geq m\}$. $\square$

Lemma 3.5 (window sums; generalising [4, Prop. 3.2(2)]). *For all $a \geq 0$ and $h \geq 1$: $c_{a+1} + c_{a+2} + \dots + c_{a+h} \leq n + T_{h-1}$.*

Proof. Let $\mu = \lambda^{(a)}$; its part-count sequence is $(c_{a+1}, c_{a+2}, \dots)$. Applying (2) to μ , for $1 \leq g \leq h$ the number c_{a+g} equals $\#\{m : \mu_m \geq g\}$ plus the number of piles created in the first $g - 1$ moves starting from μ that are still present, which is at most $g - 1$. Summing, $\sum_{g=1}^h c_{a+g} \leq \sum_{g \geq 1} \#\{m : \mu_m \geq g\} + \sum_{g=1}^h (g - 1) = |\mu| + T_{h-1} = n + T_{h-1}$. $\square$

4 Ramps

The following lemmas are Lemmas 3.4–3.6 and Proposition 3.2(3) of Griggs–Ho [4]; we give complete proofs (the proof of [4, Lemma 3.5] ends with “continue this process”, which we replace by an explicit induction) and we add the equality statement in Proposition 4.5.

Definition 4.1. A *ramp* of λ is a triple (p, q, x) of integers with $p \geq 1$, $q \geq p + 2$ and

$$c_p = x - 1, \quad c_{p+1} = c_{p+2} = \cdots = c_{q-1} = x, \quad c_q = x + 1.$$

Its *length* is $q - p$ and its *value* is x (note $x \geq 2$, since $c_p \geq 1$).

Lemma 4.2 (sandwich). *If $1 \leq i < j$ and $c_i < x < c_j$ for an integer x , then there is a ramp (p, q, x) with $i \leq p < q \leq j$.*

Proof. Let p be the largest m with $i \leq m < j$ and $c_m \leq x - 1$ (it exists: $m = i$ qualifies). By Lemma 3.2(b), $c_{p+1} \leq c_p + 1 \leq x$. Since $c_j > x$, this rules out $p + 1 = j$; so $p + 1 < j$, and $c_{p+1} \geq x$ by maximality of p . Hence $c_p = x - 1$ and $c_{p+1} = x$. Let q be the least m with $p < m \leq j$ and $c_m \geq x + 1$ (it exists: $m = j$ qualifies). Then $q \geq p + 2$ since $c_{p+1} = x$. For $p < m < q$ we have $c_m \leq x$ (minimality of q) and $c_m \geq x$ (maximality of p , as $m < j$), so $c_m = x$. Finally $x + 1 \leq c_q \leq c_{q-1} + 1 = x + 1$. $\square$

Lemma 4.3 (short ramps [4, Lemma 3.6]). *If $(p, p + 2, x)$ is a ramp, then $p \leq x$.*

Proof. Suppose $p \geq x + 1$; then $p \geq 3$. Since $|R_p| = c_p = x - 1 \leq p - 2$, not all of $N_1, \dots, N_{p-1}$ lie in R_p , while $N_{p-1} \in R_p$ (Lemma 3.2(a)). Let j be the largest index in $[1, p - 2]$ with $N_j \notin R_p$; then $N_{j+1} \in R_p$. As $c_{p+1} = c_p + 1$ and $c_{p+2} = c_{p+1} + 1$, Lemma 3.2(c) gives $R_p \subseteq R_{p+1} \subseteq R_{p+2}$, so $N_{j+1} \in R_{p+1}$ and $N_{j+1} \in R_{p+2}$. But Lemma 3.3 (with $i = p$) gives $N_{j+1} \notin R_{p+2}$, a contradiction. $\square$

Lemma 4.4 (descent [4, Lemma 3.5]). *Let (p, q, x) be a ramp with $q - p \geq 3$ and $p \geq x + 1$. Then there is a ramp (p', q', x') with*

$$p - x \leq p' \leq p - 2, \quad x' = p - p' \leq x, \quad 2 \leq q' - p' \leq q - p - 1.$$

Proof. The x labels $N_{p-x}, \dots, N_{p-1}$ exist since $p - x \geq 1$. As $|R_p| = x - 1$, one of them is not in R_p ; $N_{p-1} \in R_p$. Let p' be the largest index in $[p - x, p - 2]$ with $N_{p'} \notin R_p$ and put $y = p - p'$, so $2 \leq y \leq x$ and $N_{p'+1}, \dots, N_{p-1} \in R_p$. As $c_{p+1} = c_p + 1$, $R_p \subseteq R_{p+1}$, and with Lemma 3.2(a), $N_{p'+1}, \dots, N_p \in R_{p+1}$. Lemma 3.3 ($j = p'$, $i = p$) gives $c_{p'} = y - 1$ and $c_{p'+1} = y$.

For $1 \leq l \leq q - p - 1$ let $H(l)$ be the statement

$$H(l) : \quad c_{p'+1} = \cdots = c_{p'+l} = y \quad \text{and} \quad N_{p'+l}, N_{p'+l+1}, \dots, N_{p+l-1} \in R_{p+l}.$$

We just proved $H(1)$. Assume $H(l)$. Since $c_{p'+l} = y = p - p'$, (2) gives, for $i > p' + l$: $N_{p'+l} \in R_i \iff i \leq p + l$. Hence $N_{p'+l} \in Z_{p+l}$.

- If $l = q - p - 1$, then $p + l = q - 1$ and $c_q = c_{q-1} + 1$, so $Z_{q-1} = \emptyset$ by Lemma 3.2(c): a contradiction. So $H(q - p - 1)$ is false.
- If $l \leq q - p - 2$, then $p + l$ and $p + l + 1$ lie in $[p + 1, q - 1]$, so $c_{p+l+1} = c_{p+l} = x$ and $|Z_{p+l}| = 1$, i.e. $Z_{p+l} = \{N_{p'+l}\}$. By Lemma 3.2(b), $R_{p+l+1} = (R_{p+l} \setminus \{N_{p'+l}\}) \cup \{N_{p+l}\} \supseteq \{N_{p'+l+1}, \dots, N_{p+l}\}$. In particular $N_{p'+l+1} \in R_{p+l+1}$, i.e. $c_{p'+l+1} \geq (p + l + 1) - (p' + l + 1) = y$, and $c_{p'+l+1} \leq c_{p'+l} + 1 = y + 1$. So either $c_{p'+l+1} = y + 1$, or $c_{p'+l+1} = y$ and then $H(l + 1)$ holds.

Let l_0 be the largest $l \leq q - p - 1$ such that $H(1), \dots, H(l)$ all hold; then $1 \leq l_0 \leq q - p - 2$, and since $H(l_0 + 1)$ fails, the second item gives $c_{p'+l_0+1} = y + 1$. Thus $(p', p' + l_0 + 1, y)$ is a ramp of length $l_0 + 1 \in [2, q - p - 1]$, and $x' = y$ has the stated properties. $\square$

Proposition 4.5 (ramp bound). *Every ramp (p, q, x) satisfies $p \leq (q - p - 1)x$. If equality holds, then $(jx, jx + j + 1, x)$ is a ramp for every $j = 1, 2, \dots, q - p - 1$.*

Proof. Strong induction on the length $L = q - p \geq 2$. If $L = 2$, Lemma 4.3 gives $p \leq x$; if $p = x$, the ramp itself is $(x, x + 2, x)$. Let $L \geq 3$. If $p \leq x$ then $p \leq x < (L - 1)x$. Otherwise Lemma 4.4 provides a ramp (p', q', x') with $p' \geq p - x$, $x' \leq x$ and length $L' \leq L - 1$; by induction $p' \leq (L' - 1)x'$, hence

$$p \leq p' + x \leq (L' - 1)x' + x \leq (L - 2)x + x = (L - 1)x.$$

If $p = (L - 1)x$, all these inequalities are equalities: $p' = p - x = (L - 2)x$ and $(L' - 1)x' = (L - 2)x$; since $1 \leq L' - 1 \leq L - 2$ and $x' \leq x$, this forces $L' = L - 1$ and $x' = x$. So $((L - 2)x, (L - 2)x + L - 1, x)$ is a ramp attaining equality, and by induction $(jx, jx + j + 1, x)$ is a ramp for $j = 1, \dots, L - 2$; for $j = L - 1$ it is (p, q, x) itself. $\square$

Lemma 4.6 (long ramps [4, Lemma 3.4]). *If (p, q, x) is a ramp with $q - p = x + 1$, then $q \leq n + 1$. (Recall that $x \geq 2$ for every ramp.)*

Proof. For $p + 1 \leq j \leq q - 1$ we have $c_j = x \geq q - j$, so $N_j \in R_q$: these are x piles. Also $c_p = x - 1 < q - p$, so $N_p \notin R_q$. Since $|R_q| = c_q = x + 1$, R_q contains exactly one further label P , and $P = O_m$ for some m , or $P = N_j$ with $1 \leq j \leq p - 1$. If $P = O_m$, then $q \leq \lambda_m \leq n$. If $P = N_1$, then $c_1 \geq q - 1$, and $c_1 = \ell(\lambda) \leq n$. Suppose $P = N_j$ with $2 \leq j \leq p - 1$. Then $N_{j-1} \notin R_q$. As $c_q = c_{q-1} + 1$, $R_{q-1} \subseteq R_q$, so $N_{j-1} \notin R_{q-1}$. Next, $q - 2 \geq p + 1$ (as $x \geq 2$), so $c_{q-1} = c_{q-2} = x$ and $|Z_{q-2}| = 1$; since $c_p = x - 1 = (q - 2) - p$, $N_p \in R_{q-2} \setminus R_{q-1}$, so $Z_{q-2} = \{N_p\}$. As $j - 1 \neq p$, $N_{j-1} \in R_{q-2}$ would give $N_{j-1} \in R_{q-1}$ (Lemma 3.2(b)); hence $N_{j-1} \notin R_{q-2}$, i.e. $c_{j-1} \leq q - j - 2$. But $N_j \in R_q$ gives $c_j \geq q - j$, so $c_j \geq c_{j-1} + 2$, contradicting Lemma 3.2(b). $\square$

5 The tail lemma

This is [4, Lemma 3.3(2) and Lemma 4.3(2)], in a form that serves both the triangular and the non-triangular case.

Lemma 5.1. *Let $k \geq 2$ and $t \geq k + 1$ with $c_t = k - 1$, $c_{t+1} = k$, and $c_j \geq k - 1$ for $t \leq j \leq t + k - 2$. Then at least one of the following holds.*

- (A) *There is a ramp (p, q, k) with $t - k \leq p < q \leq t - 1$.*
- (B) *There is a ramp $(p, q, k - 1)$ with $t - k + 1 \leq p < q \leq t + 1$.*
- (C) *Every part of $\lambda^{(t-1)}$ is at most k , and $c_{t+m} = c_{t-k+m}$ for $0 \leq m \leq k - 1$.*

Proof. The k labels $N_{t-k}, \dots, N_{t-1}$ exist ($t - k \geq 1$); as $|R_t| = k - 1$, one of them is not in R_t , and $N_{t-1} \in R_t$. Let i be the largest index in $[t - k, t - 2]$ with $N_i \notin R_t$. Then $N_{i+1}, \dots, N_{t-1} \in R_t$; as $c_{t+1} = c_t + 1$, $R_t \subseteq R_{t+1}$, so $N_{i+1} \in R_{t+1}$, and Lemma 3.3 gives $c_i = t - i - 1$ and $c_{i+1} = t - i$.

Case 1: $i \geq t - k + 1$. Then $c_i \leq k - 2 < k - 1 < k = c_{t+1}$, and Lemma 4.2 gives a ramp $(p, q, k - 1)$ with $i \leq p < q \leq t + 1$: (B) holds.

Case 2: $i = t - k$. Then $c_{t-k} = k - 1$, $c_{t-k+1} = k$, and $R_t = \{N_{t-k+1}, \dots, N_{t-1}\}$ (these $k - 1$ labels lie in R_t and $|R_t| = k - 1$). If $c_j \leq k - 2$ for some $j \in [t - k + 2, t - 1]$, Lemma 4.2 applied

to $j < t + 1$ gives (B). If $c_j \geq k + 1$ for some $j \in [t - k + 2, t - 1]$, Lemma 4.2 applied to $t - k < j$ (with $c_{t-k} = k - 1 < k < c_j$) gives a ramp (p, q, k) with $t - k \leq p < q \leq j \leq t - 1$: (A) holds. Otherwise $c_j \in \{k - 1, k\}$ for all $j \in [t - k + 1, t - 1]$; we show (C). The parts of $\lambda^{(t-1)}$ are the sizes $c_j - (t - 1 - j) \leq k$ of the piles $N_j \in R_t$. Write $c_{t-1-u} = k - 1 + \varepsilon_u$ with $\varepsilon_u \in \{0, 1\}$ ($0 \leq u \leq k - 2$). Fix $m \in [0, k - 1]$ and determine R_{t+m} :

- labels not in R_t (all O 's and all N_j with $j \leq t - k$) are not in R_{t+m} (Lemma 3.2(d));
- for $t \leq j \leq t + m - 1$: $c_j \geq k - 1 \geq m \geq t + m - j$, so $N_j \in R_{t+m}$ (m labels);
- $N_{t-1-u} \in R_{t+m} \iff k - 1 + \varepsilon_u \geq m + 1 + u \iff u \leq k - 2 - m + \varepsilon_u$.

For $m = 0$ this gives $c_t = k - 1 = c_{t-k}$. For $1 \leq m \leq k - 1$ the labels N_{t-1-u} with $u \leq k - 2 - m$ are in R_{t+m} , the one with $u = k - 1 - m$ is in iff $\varepsilon_{k-1-m} = 1$, and those with $u \geq k - m$ are not. So $c_{t+m} = m + (k - 1 - m) + \varepsilon_{k-1-m} = k - 1 + \varepsilon_{k-1-m} = c_{t-k+m}$. $\square$

6 The lower bound: an explicit extremal partition

For $w \geq 1$ and $1 \leq j \leq w$ write $j \oplus_w 1 = j + 1$ if $j < w$ and $w \oplus_w 1 = 1$. For $k \geq 2$, $h \in \{1, \dots, k\}$ and $x \in \{1, \dots, k + 1\}$ put

$$X(h, x) = \Delta_{k-1} \cup (D_k \setminus \{d_k(h)\}) \cup \{d_{k+1}(x)\}.$$

Lemma 6.1. *Let $k \geq 2$ and let μ be a partition with $Y(\mu) = X(h, x)$. If $(h, x) \neq (k, 1)$, then $Y(\mathcal{B}(\mu)) = X(h \oplus_k 1, x \oplus_{k+1} 1)$. If $(h, x) = (k, 1)$, then $\mu = (k + 1, k - 1, k - 2, \dots, 2)$ and $\mathcal{B}(\mu) = \delta_k$.*

Proof. The cells of $X(h, x)$ in row 1 are among the $(1, w) = d_w(w)$: those with $w \leq k - 1$; $d_k(k)$ iff $h \neq k$; $d_{k+1}(k + 1)$ iff $x = k + 1$. So $\ell(\mu) = k - 1 + [h \neq k] + [x = k + 1]$. Likewise, from the cells $d_w(1)$ in column 1, $\mu_1 = k - 1 + [h \neq 1] + [x = 1]$. If $\ell(\mu) \leq \mu_1 - 2$, then $[h \neq k] + [x = k + 1] + 2 \leq [h \neq 1] + [x = 1] \leq 2$. This forces $h = k$ and $x = 1$. Hence for $(h, x) \neq (k, 1)$ we have $\ell(\mu) \geq \mu_1 - 1$, and Corollary 2.4 and Lemma 2.1 give $Y(\mathcal{B}(\mu)) = \rho(X(h, x)) = \Delta_{k-1} \cup (D_k \setminus \{d_k(h \oplus_k 1)\}) \cup \{d_{k+1}(x \oplus_{k+1} 1)\}$. If $(h, x) = (k, 1)$, then column 1 has $(k - 1) + 1 + 1 = k + 1$ cells. Column j with $2 \leq j \leq k - 1$ has $(k - j) + 1$ cells. Column k has none, since $d_k(k)$ is missing and Δ_{k-1} has no cell there, and there are no further columns. So $\mu = (k + 1, k - 1, \dots, 2)$, $\ell(\mu) = k - 1$, and $\mathcal{B}(\mu)$ has the parts $k - 1$ and $k, k - 2, k - 3, \dots, 1$, i.e. $\mathcal{B}(\mu) = \delta_k$. $\square$

Theorem 6.2 ([4, Thm. 3.1]). *For $k \geq 2$, $\lambda^{[k]} = (k - 1, k - 1, k - 2, \dots, 2, 1, 1)$ is a partition of T_k with $d_{\mathcal{B}}(\lambda^{[k]}) = k^2 - k$. Hence $D_{\mathcal{B}}(T_k) \geq k^2 - k$.*

Proof. The sequence is non-increasing, and its sum is $(k - 1) + T_{k-1} + 1 = T_k$. Column 1 of $Y(\lambda^{[k]})$ has the $k - 1$ cells of Δ_{k-1} . Column j ($2 \leq j \leq k$) has the $k - j$ cells of Δ_{k-1} plus $d_k(j)$. Column $k + 1$ is $\{(1, k + 1)\} = \{d_{k+1}(k + 1)\}$. So $Y(\lambda^{[k]}) = X(1, k + 1)$.

For $m \geq 0$ let $h_m \in \{1, \dots, k\}$ with $h_m \equiv 1 + m \pmod{k}$ and $x_m \in \{1, \dots, k + 1\}$ with $x_m \equiv m \pmod{k + 1}$; so $(h_0, x_0) = (1, k + 1)$ and $(h_{m+1}, x_{m+1}) = (h_m \oplus_k 1, x_m \oplus_{k+1} 1)$. We have $(h_m, x_m) = (k, 1)$ iff $m \equiv -1 \pmod{k}$ and $m \equiv 1 \pmod{k + 1}$. The number $M = k^2 - k - 1 = (k + 1)(k - 2) + 1$ satisfies both congruences, so by the Chinese remainder theorem ($\gcd(k, k + 1) = 1$) the solutions are exactly the $m \equiv M \pmod{k(k + 1)}$, and M is the least non-negative one.

By Lemma 6.1 and induction, $Y(\mathcal{B}^m(\lambda^{[k]})) = X(h_m, x_m)$ for $0 \leq m \leq M$. Then $\mathcal{B}^M(\lambda^{[k]}) = (k + 1, k - 1, \dots, 2)$ and $\mathcal{B}^{M+1}(\lambda^{[k]}) = \delta_k$, which is cyclic. For $m \leq M$, $Y(\mathcal{B}^m(\lambda^{[k]}))$ contains a cell of D_{k+1} while $Y(\delta_k) = \Delta_k$ does not. So $\mathcal{B}^m(\lambda^{[k]}) \neq \delta_k$; hence $\mathcal{B}^m(\lambda^{[k]})$ is not cyclic, by Corollary 2.6(a). Hence $d_{\mathcal{B}}(\lambda^{[k]}) = M + 1 = k^2 - k$. $\square$

7 The upper bound

Lemma 7.1 (entry, triangular case). *Let $k \geq 2$, $\lambda \vdash T_k$ and $d = d_{\mathcal{B}}(\lambda) \geq 1$. Then $\lambda^{(d-1)} = (k+1, k-1, k-2, \dots, 2)$; hence $c_d = k-1$ and $c_i = k$ for all $i \geq d+1$.*

Proof. By Corollary 2.6(a), $\lambda^{(d)} = \delta_k$, and $\mu = \lambda^{(d-1)} \neq \delta_k$ by minimality of d . Let $s = \ell(\mu)$. By definition of $\mathcal{B}$, the parts of $\delta_k = \mathcal{B}(\mu)$ are s together with the $\mu_m - 1 \geq 1$ ($\mu_m \geq 2$). So $s \in \{1, \dots, k\}$, and the other $k-1$ parts of δ_k are the numbers $\mu_m - 1$ with $\mu_m \geq 2$; hence $s \geq k-1$. If $s = k$, the numbers $\mu_m - 1$ ($\mu_m \geq 2$) are $1, \dots, k-1$ and exactly one part of μ equals 1, so $\mu = \delta_k$, which is excluded. Hence $s = k-1$, and the $k-1$ parts of μ are $z+1$ for $z \in \{1, \dots, k\} \setminus \{k-1\}$, i.e. $\mu = (k+1, k-1, \dots, 2)$. Then $c_d = \ell(\mu) = k-1$ and $c_i = \ell(\delta_k) = k$ for $i \geq d+1$. $\square$

Theorem 7.2 (Igusa, Etienne; proof of [4, Thm. 3.7]). *For every $k \geq 1$ and $\lambda \vdash T_k$, $d_{\mathcal{B}}(\lambda) \leq k^2 - k$.*

Proof. $k = 1$: the only partition of 1 is $(1) = \delta_1$, which is fixed. $k = 2$: $\mathcal{B}(1, 1, 1) = (3)$, $\mathcal{B}(3) = (2, 1)$ and $\mathcal{B}(2, 1) = (2, 1)$, so all $d_{\mathcal{B}} \leq 2 = k^2 - k$. Let $k \geq 3$ and $d = d_{\mathcal{B}}(\lambda)$. If $d \leq k$ we are done ($k \leq k^2 - k$). Assume $d \geq k+1$ and put $t = d$. By Lemma 7.1, $c_t = k-1$ and $c_j = k$ for $j \geq t+1$, so Lemma 5.1 applies.

- (C) is impossible: $\lambda^{(t-1)} = (k+1, \dots, 2)$ has the part $k+1 > k$.
- (A): a ramp (p, q, k) with $t - k \leq p < q \leq t - 1$; its length is $L = q - p \leq k - 1$. By Proposition 4.5, $p \leq (L-1)k \leq (k-2)k$, so $d = t \leq p + k \leq k^2 - k$.
- (B): a ramp $(p, q, k-1)$ with $t - k + 1 \leq p < q \leq t + 1$, of length $L \leq k$. If $L \leq k-1$, then $p \leq (k-2)(k-1)$ and $d = t \leq p + k - 1 \leq (k-1)^2 < k^2 - k$. If $L = k = (k-1) + 1$, Lemma 4.6 (with $x = k-1 \geq 2$) gives $q \leq n + 1$; as $q = p + k \geq t + 1$, we get $d = t \leq n = T_k \leq k^2 - k$, because $k \geq 3$.

$\square$

Theorem 7.3. $D_{\mathcal{B}}(T_k) = k^2 - k$ for all $k \geq 1$.

Proof. Theorem 7.2; the lower bound is Theorem 6.2 for $k \geq 2$ and trivial for $k = 1$. $\square$

8 All extremal partitions of T_k

Theorem 8.1. *Let $k \geq 4$ and $\lambda \vdash T_k$ with part-count sequence (c_i) . The following are equivalent:*

- (1) $d_{\mathcal{B}}(\lambda) = k^2 - k$;
- (2) $c_{k^2-k-1} \geq k+1$, i.e. $\mathcal{B}^{k^2-k-2}(\lambda)$ has at least $k+1$ parts;
- (3) $(jk, jk+j+1, k)$ is a ramp for every $j = 1, \dots, k-2$.

Proof. (3) $\Rightarrow$ (2): for $j = k-2$ the ramp gives $c_{(k-2)k+k-1} = c_{k^2-k-1} = k+1$.

(2) $\Rightarrow$ (1): $\mathcal{B}^{k^2-k-2}(\lambda)$ has more than $k = \ell(\delta_k)$ parts, so it is not cyclic (Corollary 2.6(a)), and $d = d_{\mathcal{B}}(\lambda) \geq k^2 - k - 1$. If $d = k^2 - k - 1$, Lemma 7.1 would give $c_d = k-1$, contradicting $c_{k^2-k-1} \geq k+1$. So $d \geq k^2 - k$, and $d \leq k^2 - k$ by Theorem 7.2.

(1) $\Rightarrow$ (3): $d = k^2 - k \geq k+1$, so we are in the main case of the proof of Theorem 7.2 with $t = d$. If (B) held, that proof would give $d \leq \max\{(k-1)^2, T_k\} < k^2 - k$. This uses $k \geq 4$: $T_k < k^2 - k \iff k > 3$. (C) is impossible. Hence (A) holds, and $d \leq p + k \leq (L-1)k + k \leq k^2 - k$ must hold with equality throughout: $L = k-1$ and $p = (L-1)k$. By the equality part of Proposition 4.5, $(jk, jk+j+1, k)$ is a ramp for $j = 1, \dots, L-1 = k-2$. $\square$

Remark 8.2. Condition (2) is decidable by performing $k^2 - k - 2$ moves, but it is a dynamical description, not a closed formula for the extremal partitions. Their number grows quickly (table below), and we do not claim a closed description. For $k \leq 3$ the extremal partitions are (1), (1, 1, 1) and (2, 2, 1, 1) (exhaustive computation over the $1 + 3 + 11$ partitions of 1, 3, 6, Appendix A).

9 Computational checks

`bs.py` (exhaustive over all partitions of each $n \leq 60$, about 1.5 minutes) confirms $D_{\mathcal{B}}(T_k) = k^2 - k$ for $1 \leq k \leq 10$. It checks $d_{\mathcal{B}}(\lambda^{[k]}) = k^2 - k$ for $2 \leq k < 80$ by direct iteration. It also confirms the equivalence $(1) \Leftrightarrow (2) \Leftrightarrow (3)$ of Theorem 8.1 for $4 \leq k \leq 10$. Number of extremal partitions of T_k :

k	3	4	5	6	7	8	9	10
$\#\{\lambda \vdash T_k : d_{\mathcal{B}}(\lambda) = k^2 - k\}$	1	3	16	65	293	1267	5686	25475

None of these computations is used in the proofs of Theorems 6.2–8.1. The only computational statement is the list for $k \leq 3$ in Remark 8.2 (Appendix A).

A Code for the exhaustive small cases

File `p3_small_cases.py` (Python 3, no dependencies; runs in well under one second). It computes $d_{\mathcal{B}}$ directly from the definition: the orbit of λ is followed until a partition repeats, and the index of the first occurrence of that partition is $d_{\mathcal{B}}(\lambda)$.

```
#!/usr/bin/env python3
"""Exhaustive small cases used in p3_c2/p3_c3/p3_c4/p3_c5.
Straight from the definitions: B(lambda), d_B(lambda) = number of steps before the
first partition that recurs in the orbit, D_B(n) = max over all partitions of n.
Runtime: well under one second."""

def partitions(n, maxp=None):
    maxp = n if maxp is None else maxp
    if n == 0:
        yield ()
        return
    for p in range(min(n, maxp), 0, -1):
        for rest in partitions(n - p, p):
            yield (p,) + rest

def B(lam):
    return tuple(sorted([len(lam)] + [x - 1 for x in lam if x > 1], reverse=True))

def d_B(lam):
    first = {}
    i = 0
    while lam not in first:
        first[lam] = i
        lam = B(lam)
        i += 1
    return first[lam] # the tail length: B^i(lam) is cyclic iff i >= this

def fmt(lam):
```

```

# compact notation: (3,3,2,1) -> 3^2 2 1
out = []
for v in sorted(set(lam), reverse=True):
    m = lam.count(v)
    out.append(f"{v}^{m}" if m > 1 else f"{v}")
return "(" + " ".join(out) + ")"

for n in [1, 3, 4, 5, 6, 7, 8, 9, 12, 17]:
    parts = list(partitions(n))
    d = {lam: d_B(lam) for lam in parts}
    D = max(d.values())
    ext = [lam for lam in parts if d[lam] == D]
    print(f"n={n}: {len(parts)} partitions, D_B={D}, extremal: {'', '.join(fmt(l) for l in ext)}")

```

Output:

```

n=1: 1 partitions, D_B=0, extremal: (1)
n=3: 3 partitions, D_B=2, extremal: (1^3)
n=4: 5 partitions, D_B=2, extremal: (1^4)
n=5: 7 partitions, D_B=3, extremal: (1^5)
n=6: 11 partitions, D_B=6, extremal: (2^2 1^2)
n=7: 15 partitions, D_B=4, extremal: (1^7)
n=8: 22 partitions, D_B=5, extremal: (1^8)
n=9: 30 partitions, D_B=7, extremal: (3 2^2 1^2)
n=12: 77 partitions, D_B=8, extremal: (3^2 2^2 1^2), (1^12)
n=17: 297 partitions, D_B=12, extremal: (1^17)

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References

- [1] E. Akin and M. Davis, Bulgarian solitaire, *Amer. Math. Monthly* 92 (1985), no. 4, 237–250.
- [2] J. Brandt, Cycles of partitions, *Proc. Amer. Math. Soc.* 85 (1982), 483–486.
- [3] G. Etienne, Tableaux de Young et solitaire bulgare, *J. Combin. Theory Ser. A* 58 (1991), 181–197.
- [4] J. R. Griggs and C.-C. Ho, The cycling of partitions and compositions under repeated shifts, *Adv. in Appl. Math.* 21 (1998), no. 2, 205–227. Preprint (dated March 2, 1998): <https://people.math.sc.edu/griggs/cycling.pdf>. Theorem and lemma numbers refer to this preprint.
- [5] K. Igusa, Solution of the Bulgarian solitaire conjecture, *Math. Mag.* 58 (1985), 259–271.
- [6] OEIS Foundation, The On-Line Encyclopedia of Integer Sequences, sequence A037306.